



# GUJARAT TECHNOLOGICAL UNIVERSITY

**Program Name: Bachelor of Engineering**

**Level: UG**

**Subject Code: BE04000261**

**Subject Name: Discrete Mathematics and Graph Theory**

|                         |                      |
|-------------------------|----------------------|
| w. e. f. Academic Year: | 2024-25              |
| Semester:               | 4                    |
| Category of the Course: | Basic Science Course |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisite:</b> | Basic Algebra                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Rationale:</b>    | This course introduces the basic concepts of discrete mathematics in the field of computer science. It covers sets, logic, functions, relations, graph theory and algebraic structures. These basic concepts of sets, logic, functions, relations and graph theory are applied to Boolean Algebra and logic networks, while the advanced concepts of functions and algebraic structures are applied to finite state machines and coding theory. |

## Course Outcomes:

| Sr. No. | CO statement                                                                                                                                                                                                                                                                                                                                                                | Marks % Weightage |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| CO-1    | Write an argument using logical notation and determine if the argument is or is not valid. To simplify and evaluate basic logic statements including compound statements, implications, inverses, converses, and contra positives using truth tables and the properties of logic. To express a logic sentence in terms of predicates, quantifiers, and logical connectives. | 18%               |
| CO-2    | Understand the basic principles of sets and operations in sets and apply counting principles to determine probabilities. Be familiar with recurrence relations.                                                                                                                                                                                                             | 11%               |
| CO-3    | Understand the basic concept of functions like domain and range of a function, identify one-to-one function, perform the composition of functions and apply the properties of functions to application problems. Use the properties of algebraic structures and be familiar with algebraic systems, groups, subgroups, Rings and Fields.                                    | 22%               |
| CO-4    | Apply relations and to determine their properties. Understand partial orders with Hasse diagrams, and the fundamental properties and types of lattices to analyze mathematical structures.                                                                                                                                                                                  | 22%               |
| CO-5    | Interpret different traversal methods for trees and graphs. Model problems in Computer Science using graphs and trees.                                                                                                                                                                                                                                                      | 27%               |

## Teaching and Examination Scheme:

| Teaching / Learning Scheme<br>(in Hours per semester) |    |    |     |                                | Total Credits | Assessment Pattern and Marks |             |                      |         |         | Total Marks |
|-------------------------------------------------------|----|----|-----|--------------------------------|---------------|------------------------------|-------------|----------------------|---------|---------|-------------|
| L                                                     | T  | P  | PBL | Total no of hours per semester |               | Theory                       |             | Tutorial / Practical |         |         |             |
|                                                       |    |    |     |                                |               | ESE (E)                      | PA / CA (M) | PA/ CA (I)           | PBL (I) | ESE (V) |             |
| 45                                                    | 30 | 00 | 15  | 90                             | 3             | 70                           | 30          | 0                    | 30      | 0       | 130         |

**Problem Based Learning (PBL)** aims to accommodate learning beyond syllabus as per clause 9.4 of NBA manual.

## Course Content:



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| Sr. No. | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Total Hrs | % of Weightage |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|
| 1       | <p><b>Propositional Logic:</b> Definition, statements &amp; notation, truth values, connectives, statement formulas &amp; truth tables, well-formed formulas, tautologies, equivalence of formulas, duality law, tautological implications, examples.</p> <p><b>Predicate Logic:</b> Definition of predicates, statement functions, variables, quantifiers, predicate formulas, free &amp; bound variables, the universe of discourse, examples, valid formulas &amp; equivalences, examples</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 08        | 18             |
| 2       | <p><b>Set Theory:</b> Basic concepts of set theory like inclusion, complement, intersection, union, Cartesian product, power set.</p> <p><b>Counting and Combinatorics:</b> Basic counting principles, permutations and combinations, binomial coefficients and the binomial theorem, Pigeonhole principle, inclusion-exclusion principle.</p> <p><b>Recurrence relations:</b> Definitions and simple examples only.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 05        | 11             |
| 3       | <p><b>Functions:</b> Basic concepts of functions like domain, range, surjective, injective, bijective, composition of functions, inverse of functions.</p> <p><b>Algebraic Structures:</b> Algebraic structures with one binary operation- semigroup, monoid, group, subgroup, normal subgroup, coset, homomorphic subgroups, Lagrange's theorem, congruence relation and quotient structures. Algebraic structures with two binary operation- Ring, integral domain and field (definitions and simple examples only).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10        | 22             |
| 4       | <p><b>Relations:</b> Definition, binary relation, representation, domain, range, universal relation, void relation, union, intersection, and complement operations on relations, properties of binary relations in a set: reflexive, symmetric, transitive, anti-symmetric relations, partition and covering of a set, equivalence relation, equivalence classes, compatibility relation, maximum compatibility block, composite relation, converse of a relation, transitive closure of a relation R in set X</p> <p><b>Partial Ordering:</b> Definition, examples, simple or linear ordering, totally ordered set (chain), frequently used partially ordered relations, representation of partially ordered sets, Hasse diagrams, least &amp; greatest elements, minimal &amp; maximal elements, least upper bound (Supremum), greatest lower bound (Infimum), Well-ordered partially ordered sets (Posets). Lattice as Posets, complete, distributive, modular and complemented lattices, Boolean and pseudo Boolean lattices (definitions and simple examples only).</p> | 10        | 22             |
| 5       | <p><b>Graphs:</b> Introduction, definition, examples; Nodes, edges, adjacent nodes, directed and undirected edge, directed graph, undirected graph, examples; Initiating and terminating nodes, loop (sling), distinct edges, parallel edges, multi-graph, simple graph, weighted graphs, examples; Isolated nodes, null graph, isomorphic graphs, examples; Degree, In-degree, out-degree, total degree of a node, examples; Subgraphs: definition, examples; Converse (reversal or directional dual) of a digraph, examples; Path: Definition, paths of a given graph, length of path, examples; Simple path (edge simple), elementary path (node simple), examples; Cycle (circuit), elementary cycle, examples; <b>Reachability:</b> Definition, geodesic, distance, examples; Properties of reachability, the triangle inequality; Reachable set of a given node, examples;</p>                                                                                                                                                                                         | 12        | 27             |



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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Node base, examples<br><b>Connectedness:</b> Definition, weakly connected, strongly connected, unilaterally connected, examples; Strong, weak, and unilateral components of a graph, examples; Applications to represent resource allocation status of an operating system, and detection and correction of deadlocks; Matrix representation of graph: Definition, adjacency matrix, Boolean (or bit) matrix, examples; Determine number of paths of length n through adjacency matrix, examples; Path (Reachability) matrix of a graph, examples; Warshall's algorithm to produce path matrix, Flowchart.<br><b>Trees:</b> Definition, branch nodes, leaf (terminal) nodes, root, examples; Different representations of a tree, examples; Binary tree, m-ary tree, full (or complete) binary tree, examples; Converting any m-ary tree to a binary tree, examples; Representation of a binary tree: Linked-list; Tree traversal: Pre-order, in-order, post-order traversal, examples; Algorithms; Applications of List structures and graphs |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

## Suggested Specification table with Marks (Theory): (For B.E. only)

| Distribution of Theory Marks |         |         |         |         |         |
|------------------------------|---------|---------|---------|---------|---------|
| R Level                      | U Level | A Level | N Level | E Level | C Level |
| 30                           | 40      | 20      | 10      | 00      | 00      |

**R: Remembrance; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create and above Levels (Revised Bloom's Taxonomy)**

**Note:** This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary slightly from above table.

## Reference Books:

1. J. P. Tremblay and R. Manohar, Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw-Hill, 1997.
2. S. Lipschutz and M. L. Lipson, Schaum's Outline of Theory and Problems of Discrete Mathematics, 2nd Ed., Tata McGraw-Hill, 1999.
3. K. H. Rosen, Discrete Mathematics and its applications, Tata McGraw-Hill, 6th Ed., 2007.
4. David Liben-Nowell, Discrete Mathematics for Computer Science, Wiley publication, July 2017.
5. Eric Gossett, Discrete Mathematics with Proof, 2nd Edition, Wiley publication, July 2009.
6. Josheph A. Gallian, Contemporary Abstract Algebra, Chapman & Hall, 11th edition, July 2025.

**Suggested MOOC Courses:** [https://onlinecourses.nptel.ac.in/noc26\\_cs68/preview](https://onlinecourses.nptel.ac.in/noc26_cs68/preview)

## List of Tutorials:

| Sr. No | Tutorials | CO |
|--------|-----------|----|
|--------|-----------|----|



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|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1 | <b>Title: Propositional logic: Truth values, truth tables &amp; tautology</b><br><b>Objective:</b> To understand and apply the concepts of propositional logic, including Boolean formulas, generating truth tables, and determining a formula is a tautology, contradiction, or contingent.<br><b>Procedure:</b> Learn the foundation of propositional logic: truth values, truth tables and tautology through guided examples in the tutorials                                                                         | 1 |
| 2 | <b>Title: Predicate logic evaluation over finite domain</b><br><b>Objective:</b> To study and evaluate quantified formulas over a finite universe, and to clearly distinguish between free and bound variables.<br><b>Procedure:</b> Learn the Predicate logic evaluation over finite domain through examples, definitions, formula evaluation, and application-oriented exercises during the tutorial.                                                                                                                  | 1 |
| 3 | <b>Title: Set operations and verifying identities</b><br><b>Objective:</b> To understand fundamental set concepts and perform operations such as union, intersection, difference, symmetric difference, and power set construction.<br><b>Procedure:</b> Learn about the universal set, types of sets, and standard set operations (union, intersection, complement, difference, symmetric difference, power set). Verify identities, including De Morgan's laws, through illustrative examples during the tutorial.     | 2 |
| 4 | <b>Title: Counting: <math>nPr</math>, <math>nCr</math>, Pascal triangle &amp; pigeonhole Principle</b><br><b>Objective:</b> To understand and apply core combinatorial concepts including $nPr$ , $nCr$ , Pascal's Triangle (up to $n=10$ ), the Pigeonhole Principle, and to explore their computational and practical uses.<br><b>Procedure:</b> Work through examples involving permutations and combinations, construct Pascal's Triangle, apply the Pigeonhole Principle, and prove related identities in tutorial. | 2 |
| 5 | <b>Title: Recurrence relations: iteration &amp; closed-form (characteristic equation)</b><br><b>Objective:</b> To compute the terms for linear homogeneous recurrences with constant coefficients by iteratively and to derive closed form solutions using the characteristic polynomial (simple roots).<br><b>Procedure:</b> Solve the recurrence relation using illustrative examples.                                                                                                                                 | 2 |
| 6 | <b>Title: Functions: mapping, composition &amp; inverse</b><br><b>Objective:</b> To understand and apply the concepts of functions, one-to-one and onto mappings, composition of functions and inverse functions.<br><b>Procedure:</b> Determine whether the given mapping is function or not, check one-to-one and onto function, find composition of functions, inverse of a function with the help of examples.                                                                                                       | 3 |
| 7 | <b>Title: Permutations &amp; group basics: cycle notation, inverse, subgroup generation</b><br><b>Objective:</b> To represent permutations, convert them to cycle notation, compute inverses, and generate groups, subgroups through repeated composition.<br><b>Procedure:</b> Work through examples involving groups—such as abelian groups, cyclic groups, permutation groups, and subgroups—to illustrate these concepts and operations.                                                                             | 3 |
| 8 | <b>Title: Normal Subgroup, Lagrange Theorem, Rings and Field</b><br><b>Objective:</b> To understand normal subgroups, Lagrange's Theorem, and the basic concepts of rings and fields.<br><b>Procedure:</b> Work through examples involving normal subgroups, Lagrange's Theorem (including a demonstrated example), and fundamental operations in rings and fields.                                                                                                                                                      | 3 |
| 9 | <b>Title: Relations: Properties and examples</b><br><b>Objective:</b> To understand and apply the concept of relation, their types, and their representations.<br><b>Procedure:</b> Understand relation, various types of relation, definitions, matrix and graphs representation of relations, properties of relations with the help of examples.                                                                                                                                                                       | 4 |



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|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 10 | <p><b>Title: Transitive closure: Warshall algorithm &amp; reachability matrix</b></p> <p><b>Objective:</b> To Implement Warshall algorithm to compute the transitive closure of a relations to understand reachability.</p> <p><b>Procedure:</b> Apply Warshall's algorithm to the adjacency matrix of a given relation in order to obtain its transitive closure, complete different examples.</p>                                                                                                                                                                                                                                                                                                                                                                            | 4 |
| 11 | <p><b>Title: Partial orders &amp; Hasse diagram generation</b></p> <p><b>Objective:</b> To understand partial orders and generate Hasse diagrams, identifying least, greatest, minimal, and maximal elements.</p> <p><b>Procedure:</b> Check whether the given relation is a partial order and draw its Hasse diagram. Identify least, greatest, minimal, maximal elements, and chains using different examples.</p>                                                                                                                                                                                                                                                                                                                                                           | 4 |
| 12 | <p><b>Title: Lattice: Distributive, Modular &amp; complemented (small finite lattices)</b></p> <p><b>Objective:</b> To Understand small finite lattice defined as Poset, and if so, whether they are distributive, modular, complemented.</p> <p><b>Procedure:</b> Check whether the given Poset is a lattice or not, and provide examples and Counter examples. Visualize each lattice using a Hasse diagram, and compute meet and join for different type of examples.</p>                                                                                                                                                                                                                                                                                                   | 4 |
| 13 | <p><b>Title: Graph basics: Adjacency matrix, degree, connected components</b></p> <p><b>Objective:</b> To understand graphs and their properties, different types of graphs, its elements, produce adjacency/incidence matrix, compute degree sequences, and find connected components (for undirected graphs) or strongly connected components (for directed graphs)</p> <p><b>Procedure:</b> Introduce different types of graphs and their elements with examples, list its properties, compute common graph parameters, and demonstrate matrix-based representations and their properties through examples.</p>                                                                                                                                                             | 5 |
| 14 | <p><b>Title: Graph algorithms: BFS/DFS, shortest path (unweighted), cycle detection</b></p> <p><b>Objective:</b> To understand and implement BFS and DFS traversals, reconstruct shortest paths in unweighted graphs, and detect cycles in both directed and undirected graphs.</p> <p><b>Procedure:</b> <b>BFS:</b> Implement BFS using a queue to compute distances and parent nodes; reconstruct shortest paths from source to target with various graph examples. <b>DFS:</b> Implement DFS and use it to detect cycles, employing visited/recursion-stack techniques for directed graphs, illustrate with examples.</p>                                                                                                                                                   | 5 |
| 15 | <p><b>Title: Weighted graphs: Dijkstra (nonnegative weights) &amp; MST (Kruskal) + tree traversals</b></p> <p><b>Objective:</b> To implement Dijkstra's algorithm for shortest paths with non-negative weights, Kruskal algorithm using DSU for MST, and practice binary tree representations with pre/in/post order traversals.</p> <p><b>Procedure:</b></p> <ol style="list-style-type: none"> <li><b>Dijkstra:</b> Implement Dijkstra's algorithm for various weighted graphs and solve illustrative examples.</li> <li><b>Kruskal:</b> Implement Kruskal's algorithm with DSU for MST on different graphs and work through examples.</li> <li><b>Binary Trees:</b> Practice binary tree representations and perform pre/in/post-order traversals with examples.</li> </ol> | 5 |



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## List of suggested activities for Problem Based Learning:

| Sr. No | Name of Activity                                                                                                                                                                          | No. of Hours                                                      | Evaluation Criteria                                                                         |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1      | Assignments                                                                                                                                                                               | 5 assignments $\times$ 1h each;<br>Total = 5h                     | Based on completeness, correctness and submission of assignments                            |
| 2      | Technical Video-Based Learning – Watch online lectures/tutorials on topics covered                                                                                                        | Duration = 4h, Report & Presentation = 1h; Total = 5h             | Report / Presentation on key learning outcomes                                              |
| 3      | Preparing Posters, Charts, Power Point Presentation or documents (using equation editor, Mathtype, Latex etc.) of all important definitions, theorems, formulae etc on topics of syllabus | Preparing, Designing and Presenting Visual Content = 5h           | Assessment based on creativity, clarity, and presentation skills                            |
| 4      | Seminar / Presentation – Study and present a technical topic beyond syllabus                                                                                                              | Study/Prep = 3h, Report = 1h, Presentation = 1h; Total = 5h       | Based on technical depth, quality of report and presentation skills                         |
| 5      | Real-World Case Study problems based on syllabus                                                                                                                                          | Study = 4h, Report prep = 1h; Total = 5h                          | Based on correctness, completeness and analysis of the report                               |
| 6      | Implementing the concepts of syllabus in programs                                                                                                                                         | Completing at least five programs = 5h                            | Review based on the implementation, results, and its output                                 |
| 7      | Modeling and Simulation on use of concept in Engineering Systems (developing models and performing simulations)                                                                           | Model formulation (4h) + Simulation and result analysis (1h) = 5h | Evaluation based on model accuracy, computational efficiency, and interpretation of results |

### Note:

- The activities listed above are suggestive, and faculty members have the flexibility to select and modify them as needed.
- The total self-learning hours remain fixed at 15 hours, ensuring comprehensive coverage of topics of Discrete Mathematics.
- Faculty can adjust the distribution of hours across different activities while maintaining a balanced learning approach.
- All records pertaining to the evaluation and assessment of self-learning activities must be properly maintained and preserved at the institute level. These records should be made available to the university upon request.
- Institutes are encouraged to utilize digital platforms, such as Microsoft Teams, for effective record keeping and to ensure transparency in the evaluation and assessment of self-learning activities.

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# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000251

Subject Name: Computer Organization & Architecture

|                        |             |
|------------------------|-------------|
| w.e.f. Academic Year:  | A.Y.2024-25 |
| Semester:              | 4           |
| Categor of the Course: | PCC         |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisite:</b> | Basic understanding of computer and programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Rationale:</b>    | This subject helps students to understand the internal architecture and working of computers. Basically, how computers evolved, how does it represent any type of data, how does it conduct arithmetic, logic, storage operations. The process computer follows when it executes a program starting from writing a program to machine codes and its execution. The understanding of these aspects of computers is really essential to understand the power as well as limitations of computers which is important to be a good computer engineer and/or scientist. |

## Course Outcome:

After Completion of the Course, Student will be able to:

| No | Course Outcomes                                                                                                                          | RBT Level |
|----|------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1  | Identify and explain the basic structure and functional units of a digital computer.                                                     | R, U      |
| 2  | Write assembly language programs and identify the role and working of various functional units of a computer for executing instructions. | R, A      |
| 3  | Design processing unit using the concepts of ALU and control logic design.                                                               | U, A      |
| 4  | Design circuits for interfacing memory and I/O with processors.                                                                          | U, A      |
| 5  | Comprehend the features and performance parameters of different types of computer architectures.                                         | R, U      |

*\*Revised Bloom's Taxonomy(RBT)*

## Teaching and Examination Scheme:

| Teaching - Learning Scheme<br>(in Hours per Semester) |   |    |      |    | Total<br>Credit<br>s =<br>TH/30 | Assessment Pattern and Marks |           |                      |         |            | Total<br>Marks |
|-------------------------------------------------------|---|----|------|----|---------------------------------|------------------------------|-----------|----------------------|---------|------------|----------------|
| L                                                     | T | P  | PBL* | TH |                                 | Theory                       |           | Tutorial / Practical |         |            |                |
|                                                       |   |    |      |    |                                 | ESE<br>(E)                   | PA<br>(M) | PA (I)               | PBL (I) | ESE<br>(V) |                |
| 45                                                    | 0 | 30 | 15   | 90 | 03                              | 70                           | 30        | 20                   | 30      | 50         | 200            |

**\* Problem Based Learning (PBL) aims to accommodate learning beyond syllabus as per clause 9.4 of NBA manual.**



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Subject Name: Computer Organization & Architecture

## Course Content:

| Unit No. | Content                                                                                                                                                                                                                                                                                     | No. of Hours | % of Weightage |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------|
| 1        | Structure of Computers: Computer types, Functional units, Basic operational concepts, Von Neumann Architecture, Bus Structures, Software, Performance, Multiprocessors and Multicomputer, Data representation, Fixed and Floating point, Error detection and correction codes.              | 2            | 5              |
| 2.       | Register Transfer and Micro-operations: Register Transfer language, Register Transfer, Bus and Memory Transfers (Tree-State Bus Buffers, Memory Transfer), Arithmetic Micro Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logical shift unit                       | 6            | 10             |
| 3.       | Basic Computer Organization and Design Instruction codes, Computer registers, computer instructions, Timing and Control, Instruction cycle, Memory-Reference Instructions, Input-output and interrupt, Complete computer description, Design of Basic computer, Design of Accumulator Unit. | 4            | 15             |
| 4.       | Assembly Language Programming Introduction, Machine Language, Assembly Language Programming: Arithmetic and logic operations, looping constructs, Subroutines, I-O Programming.                                                                                                             | 6            | 10             |
| 5.       | Micro programmed Control Organization: Control Memory, Address sequencing, Micro program example, Design of Control Unit                                                                                                                                                                    | 2            | 5              |
| 6.       | Central Processing Unit Introduction, General Register Organization, Stack Organization, Instruction format, Addressing Modes, Data transfer and manipulation, Program control, Reduced Instruction Set Computer (RISC) & Complex Instruction Set Computer (CISC)                           | 6            | 15             |
| 7.       | Pipeline And Vector Processing Flynn's taxonomy, Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction, Pipeline, RISC Pipeline, Vector Processing, Array Processors                                                                                                            | 3            | 5              |
| 8.       | Input-Output Organization Input-Output Interface, Asynchronous Data Transfer, Modes Of Transfer, Priority Interrupt, DMA, Input-Output Processor (IOP), CPU IOP Communication, Serial communication.                                                                                        | 4            | 10             |
| 9.       | Memory Organization Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory.                                                                                                                                                                      | 4            | 10             |
| 10.      | Computer Arithmetic Introduction, Addition and subtraction, Multiplication Algorithms (Booth Multiplication Algorithm), Division Algorithms, Floating Point Arithmetic operations, Decimal Arithmetic Unit.                                                                                 | 4            | 10             |
| 11.      | Multiprocessors Characteristics of Multiprocessors, Interconnection Structures, Inter-processor Arbitration, Inter-processor Communication and Synchronization, Cache Coherence, Shared Memory Multiprocessors.                                                                             | 4            | 5              |



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## Suggested Specification Table with Marks(Theory):

| Distribution of Theory Marks (in%) |        |        |        |         |        |
|------------------------------------|--------|--------|--------|---------|--------|
| RLevel                             | ULevel | ALevel | NLevel | E Level | CLevel |
| 20                                 | 40     | 20     | 20     | --      | -      |

Where R: Remember; U: Understanding; A: Application; N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

## References/Suggested Learning Resources:

### (a) Books:

1. M. Morris Mano, "Computer System Architecture", Pearson Education
2. Yale N. Patt, Sanjay J. Patel, "Introduction to Computing Systems" McGraw Hill.
3. Hamacher, Vranesic, Zaky, "Computer Organization", McGraw Hill.
4. Andrew S. Tanenbaum and Todd Austin, "Structured Computer Organization", Pearson Education
5. N. D. Jotwani, "Computer system organization", McGraw Hill
6. R.S. Gaonkar, "Microprocessor Architecture, Programming and Applications with 8085A", Penram International
7. Douglas Hall, Microprocessors and Interfacing, TMH.
8. Computer Organisation and Architecture, Smruti R. Sarangi, McGrawHill (2015)

### (b) Open source software and website:

#### NPTEL Course:

1. Computer Architecture By Prof. Smruti Ranjan Sarangi | IIT DelhA
2. Computer Architecture and Organization By Prof. Indranil Sengupta, Prof. Kamalika Datta | IIT Kharagpur
3. Computer Architecture By Prof. Anshul Kumar | IIT Delhi

## Suggested Course Practical List:(List can be change according to Latest Development)

1. Implement Booth's Algorithm
2. Write the working of 8085 simulator GNUsim8085 and basic architecture of 8085 along with small introduction.
3. Write an assembly language code in GNUsim8085 to add two 8 bit numbers
4. Write an assembly language code in GNUsim8085 to store numbers in reverse order in memory location.
5. Write an assembly language code in GNUsim8085 to implement arithmetic instruction
6. Write an assembly language code in GNUsim8085 to find the factorial of a number.
7. Write an assembly language code in GNUsim8085 to implement logical instructions.
8. Write an assembly language code in GNUsim8085 with user defined function which accepts two 8 bit numbers and return their sum.
9. Design ALU using Logisim.
10. Implement 16-bit single-cycle MIPS processor in Verilog HDL



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**Subject Name: Computer Organization & Architecture**

• **List of suggested activities for Problem Based Learning:**

| Sl. No. | Name of the activity                                                                                 | No. of hours                                                                               | Evaluation Criteria                                                                     |
|---------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 1       | Assignment writing. Numerical based assignment is preferable.                                        | 5 assignments of 3h each. Total = 15h                                                      | Based on the assignment submitted.                                                      |
| 2       | Problem solving/Coding using C, C++, Python, SCILAB, MATLAB, MS-EXCEL or any other relevant software | 5 small coding-based problems of 3h each. Total = 15h                                      | Based on the coding solution submitted.                                                 |
| 3       | Technical Video based learning related to the subject                                                | Duration of video = 5h<br>Report preparation & Presentation = 10h<br>Total = 15h           | Report /presentation based on the video learning outcomes.                              |
| 4       | Discussion on research paper based on relevant subject                                               | 3 research paper = 15h                                                                     | Summarize research paper and evaluation critical parameters                             |
| 5       | Poster/chart/power point preparation on technical topics                                             | Duration = 10 h                                                                            | Based on poster/chart preparation and presentation skills                               |
| 6       | Application/Software development                                                                     | Duration = 15 h                                                                            | Depending on the complexity of the Application/Software                                 |
| 7       | Group Discussion on emerging/trending technical topics based on subject                              | Duration = 1 h each                                                                        | Based on performance in group discussion, technical depth, knowledge etc.               |
| 8       | Seminar / Presentation                                                                               | Duration for study and preparation=5h<br>Report writing=3h<br>Presentation=2h<br>Total=10h | Topic can be selected technical content beyond syllabus                                 |
| 9       | Real world case studies-based learning                                                               | Duration of data collection/study = 5h<br>Report preparation = 10h Total = 15h             | Based on in-depth study, technical depth, data collected, fact finding, etc.            |
| 10      | Working/non-working model on technical topics                                                        | Working = 12 h<br>Non- working = 8 h                                                       | Based on inter department/external evaluation                                           |
| 11      | Self-learning on-line course                                                                         | Minimum duration of the course should be 15h.                                              | Examination based assessment at the end of course. Based on the certificate produced.   |
| 12      | Complex problem solving                                                                              | Maximum 3 problem. Study of the problem and solution finding, Total = 15h                  | Based on the depth of the solution submitted.                                           |
| 13      | Industry/Research laboratory visit                                                                   | Visit = 5h, Report preparation = 5h<br>Total = 10h                                         | Based on report submitted. Report should contain observations and calculations based on |



# GUJARAT TECHNOLOGICAL UNIVERSITY

**Program Name: Bachelor of Engineering**

**Level: UG**

**Subject Code: BE04000251**

**Subject Name: Computer Organization & Architecture**

|    |                                                                                                                            |                                                                                                                     |                                                        |
|----|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
|    |                                                                                                                            |                                                                                                                     | industry/ lab data.                                    |
| 14 | Videos on Industrial safety aspects based on subject                                                                       | Duration of video = 5h<br>Report preparation = 5h<br>Total = 10h                                                    | Based on quiz/report submitted                         |
| 15 | Industrial exposure for 2-3 days to observe and provide tentative solutions on society/environment /health/any other issue | Duration = 15 h for industrial exposure<br><br>Problem identification and tentative solution = 10 h<br>Total = 20 h | Based on evaluation of critical problems and solutions |

Note:

- All the suggested activity should be related to the subject.
- Min 3 activities must be carried out as per the availability of faculties and students.
- The number of hours is suggestive. Faculty can sub-divide the number of hours based on the activity. However, total number of hours is fixed.
- Rubrics for the evaluation can be prepared by the faculty.

\* \* \* \* \*



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000241

Subject Name: Analysis and Design of Algorithms

|                        |         |
|------------------------|---------|
| WEF Academic Year      | 2024-25 |
| Semester               | 4       |
| Category of the Course | PCC     |

|                       |                                                                                                                                                                                                                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisite :</b> | Data Structure, Programming                                                                                                                                                                                                                                                                         |
| <b>Rationale :</b>    | Developing efficient algorithms is crucial in modern computer engineering, as today's applications demand optimized performance in terms of time, space, and energy. This course provides the knowledge and skills to understand and analyze efficient algorithms for a wide range of applications. |

## Course Outcome :

After Completion of the Course, Student will able to :

| No | Course Outcomes                                                           | RBT Level* |
|----|---------------------------------------------------------------------------|------------|
| 01 | Identify Time and Space complexities of various algorithms                | RM         |
| 02 | Solve and analyze the problems divide and conquer methods                 | UN         |
| 03 | Solve and analyze the problems using greedy and dynamic programming       | AP         |
| 04 | Understand and solve graph based problems                                 | CR         |
| 05 | Apply backtracking and branch and bound methods to solve various problems | EL         |

\*RM: Remember, UN: Understand, AP: Apply, AN: Analyze, EL: Evaluate, CR: Create

## Course Scheme :

| Teaching - Learning Scheme<br>(in Hours per Semester) |   |    |      |     | Total<br>Credits<br>=<br>TH/30 | Assessment Pattern and Marks |           |                      |            |            | Total<br>Marks |
|-------------------------------------------------------|---|----|------|-----|--------------------------------|------------------------------|-----------|----------------------|------------|------------|----------------|
| L                                                     | T | P  | PBL* | TH  |                                | Theory                       |           | Tutorial / Practical |            |            |                |
|                                                       |   |    |      |     |                                | ESE<br>(E)                   | PA<br>(M) | PA<br>(I)            | PBL<br>(I) | ESE<br>(V) |                |
| 45                                                    | 0 | 30 | 45   | 120 | 04                             | 70                           | 30        | 20                   | 30         | 50         | 200            |

\* Problem Based Learning (PBL) aims to accommodate learning beyond syllabus as per clause 9.4 of NBA manual.



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000241

Subject Name: Analysis and Design of Algorithms

## Course Content:

| Sr. No. | Course Content                                                                                                                                                                                                                                                                                                             | No. of Hours | % of Weightage |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------|
| 1       | <b>Introduction and Analysis of Algorithm:</b> Introduction to algorithm, Properties of good algorithm, Average, Best and worst case analysis, Asymptotic Notations (Big Oh, Big Theta, Big Omega), Disjoint set operations, Union and find algorithms,                                                                    | 4            | 10             |
| 2       | <b>Analysis of Sorting Algorithms:</b> Comparison based sorting: Bubble sort, Selection sort, Insertion sort, Heap sort, Sorting in linear time : Bucket sort, Radix sort and Counting sort                                                                                                                                | 6            | 15             |
| 3       | <b>Divide and Conquer:</b> Introduction, General method, Recurrence and different methods to solve recurrence, Problem solving using Divide and Conquer: Binomial Coefficient, Binary Search, Max-Min problem, Merge Sort, Quick Sort, Quick sort using median of three, Matrix Multiplication, Exponential.               | 6            | 15             |
| 4       | <b>Dynamic Programming:</b> Introduction, General method, Elements of Dynamic Programming, The Principle of Optimality<br>Problem Solving using Dynamic Programming: Binomial Coefficient, Making Change Problem, Knapsack problem, Matrix chain multiplication, Longest Common Subsequence, Shortest Common Supersequence | 6            | 15             |
| 5       | <b>Greedy Algorithm:</b> Introduction, General method, Characteristics of greedy algorithms, Elements of greedy strategy<br>Problem solving using Greedy method: Making change problem, Activity selection problem, Knapsack problem (0/1 and Fractional), Job Scheduling Problem,                                         | 5            | 10             |
| 6       | <b>Graph Algorithms:</b> Introduction, Types of graph, Representation Graph, Traversing Graphs: Depth First Search, and Breath First Search, Topological sort, Strongly Connected components.                                                                                                                              | 6            | 15             |



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000241

Subject Name: Analysis and Design of Algorithms

|   |                                                                                                                                                                                                                                                                                    |           |            |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|
|   | Problems solving: Finding articulation point, Minimum Spanning trees (Kruskal's algorithm, Prim's algorithm) using greedy approach, Single pair shortest path using greedy method, All Points Shortest path using Dynamic Programming,                                             |           |            |
| 7 | <b>Backtracking:</b> Introduction, General method, n-queen problem, Sum of subset problem, Graph coloring problem, Hamiltonian cycle problem,<br><b>Branch and Bound:</b> General method, LC branch and bound, FIFO branch and bound, Traveling salesman problem, Knapsack problem | 6         | 15         |
| 8 | <b>Introduction to NP-Completeness:</b> The class P and NP, Polynomial reduction, NP- Complete Problems, NP-Hard Problems.                                                                                                                                                         | 3         | 05         |
|   | <b>Total</b>                                                                                                                                                                                                                                                                       | <b>42</b> | <b>100</b> |

## Reference Book :

- Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, PHI.
- Introduction to Design and Analysis of Algorithms, Anany Levitin, Pearson.
- Fundamental of Algorithms by Gills Brassard, Paul Bratley, PHI.
- Foundations of Algorithms, Shailesh R Sathe, Penram

## Suggested Course Practical List:

1. Perform following sorting operation and measure the execution time for sufficient large input: Selection Sort, Bubble Sort, Insertion Sort
2. Implement Linear search and Binary Search
3. Perform following sorting operation and measure the execution time for sufficient large input: Merge Sort, Quick Sort
4. Solve Given Problems using Greedy Strategy: Make a change problem, Activity Selection
5. Solve Given Problems using Greedy Strategy: Problem, 0/1 Knapsack Problem, Job Scheduling Problem
6. Solve Given Problems using Greedy Strategy: MST using Prim's Algorithm, MST using Krushkal's Algorithm
7. Solve Given Problems Using Dynamic Programming: Make a change problem, 0/1 Knapsack Problem
8. Solve Given Problems Using Dynamic Programming: Matrix Chain Multiplication



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000241

Subject Name: Analysis and Design of Algorithms

9. Solve Given Problems Using Dynamic Programming: Longest Common Subsequence (LCS), Shortest Common Super sequence (SCS)
10. Solve following problems using Backtracking 4-Queen Problem, 8-Queen Problem

## List of Laboratory/Learning Resources Required:

- <https://codecrucks.com/what-is-algorithm-nutshell-explanation/>
- <https://www.datacamp.com/blog/what-is-an-algorithm>
- <https://khyatirnimal.wordpress.com/design-analysis-and-algorithm/>
- <https://elogeel.wordpress.com/category/algorithms-design-and-analysis/>

## • List of suggested activities for Problem Based Learning:

| Sl. No. | Name of the activity                                                                                 | No. of hours                                                                     | Evaluation Criteria                                                       |
|---------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1       | Assignment writing. Numerical based assignment is preferable.                                        | 5 assignments of 3h each. Total = 15h                                            | Based on the assignment submitted.                                        |
| 2       | Problem solving/Coding using C, C++, Python, SCILAB, MATLAB, MS-EXCEL or any other relevant software | 5 small coding-based problems of 3h each. Total = 15h                            | Based on the coding solution submitted.                                   |
| 3       | Technical Video based learning related to the subject                                                | Duration of video = 5h<br>Report preparation & Presentation = 10h<br>Total = 15h | Report /presentation based on the video learning outcomes.                |
| 4       | Discussion on research paper based on relevant subject                                               | 3 research paper = 15h                                                           | Summarize research paper and evaluation critical parameters               |
| 5       | Poster/chart/power point preparation on technical topics                                             | Duration = 10 h                                                                  | Based on poster/chart preparation and presentation skills                 |
| 6       | Application/Software development                                                                     | Duration = 15 h                                                                  | Depending on the complexity of the Application/Software                   |
| 7       | Group Discussion on emerging/trending technical topics based on subject                              | Duration = 1 h each                                                              | Based on performance in group discussion, technical depth, knowledge etc. |
| 8       | Seminar / Presentation                                                                               | Duration for study and preparation=5h Report                                     | Topic can be selected technical content beyond syllabus                   |



# GUJARAT TECHNOLOGICAL UNIVERSITY

**Program Name: Bachelor of Engineering**

**Level: UG**

**Subject Code: BE04000241**

**Subject Name: Analysis and Design of Algorithms**

|    |                                                                                                                            |                                                                                                                     |                                                                                                             |
|----|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|    |                                                                                                                            | writing=3h<br>Presentation=2h<br>Total=10h                                                                          |                                                                                                             |
| 9  | Real world case studies-based learning                                                                                     | Duration of data collection/study = 5h<br>Report preparation = 10h<br>Total = 15h                                   | Based on in-depth study, technical depth, data collected, fact finding, etc.                                |
| 10 | Working/non-working model on technical topics                                                                              | Working = 12 h<br>Non- working = 8 h                                                                                | Based on inter department/external evaluation                                                               |
| 11 | Self-learning on-line course                                                                                               | Minimum duration of the course should be 15h.                                                                       | Examination based assessment at the end of course. Based on the certificate produced.                       |
| 12 | Complex problem solving                                                                                                    | Maximum 3 problem. Study of the problem and solution finding, Total = 15h                                           | Based on the depth of the solution submitted.                                                               |
| 13 | Industry/Research laboratory visit                                                                                         | Visit = 5h, Report preparation = 5h<br>Total = 10h                                                                  | Based on report submitted. Report should contain observations and calculations based on industry/ lab data. |
| 14 | Videos on Industrial safety aspects based on subject                                                                       | Duration of video = 5h<br>Report preparation = 5h<br>Total = 10h                                                    | Based on quiz/report submitted                                                                              |
| 15 | Industrial exposure for 2-3 days to observe and provide tentative solutions on society/environment /health/any other issue | Duration = 15 h for industrial exposure<br><br>Problem identification and tentative solution = 10 h<br>Total = 20 h | Based on evaluation of critical problems and solutions                                                      |

Note:

- All the suggested activity should be related to the subject.
- Min 3 activities must be carried out as per the availability of faculties and students.
- The number of hours is suggestive. Faculty can sub-divide the number of hours based on the activity. However, total number of hours is fixed.
- Rubrics for the evaluation can be prepared by the faculty.

\* \* \* \* \*



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000231

Subject Name: Object Oriented Programming

|                         |                          |
|-------------------------|--------------------------|
| w. e. f. Academic Year: | 2024-25                  |
| Semester:               | 4                        |
| Category of the Course: | Professional Core Course |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisite:</b> | Basics of Programming in C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Rationale:</b>    | The course Object-Oriented Programming using Java introduces students to the fundamental principles of object-oriented design such as encapsulation, inheritance, polymorphism, and abstraction, while providing hands-on experience in implementing them using Java. Building on their prior knowledge of procedural programming, this course helps students transition to object-oriented thinking and equips them with the skills to design modular, reusable, and maintainable applications. With Java's wide industry adoption, platform independence, and rich libraries, the syllabus ensures that learners not only gain theoretical understanding but also develop practical, industry-relevant coding skills essential for real-world software development. |

## Course Outcomes:

| Sr. No. | CO statement                                                                                                                                                                                                           | Marks% weightage |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| CO-1    | Describe the structure of Java programs and the foundational elements of the Java programming language, including data types, operators, control structures, memory model, and execution environment.                  | 10               |
| CO-2    | Apply Object-Oriented Programming principles such as encapsulation, inheritance, abstraction, and polymorphism to design modular and reusable Java applications using classes, constructors, interfaces, and packages. | 40               |
| CO-3    | Implement exception handling, multithreading, and file I/O operations to develop robust and efficient Java applications.                                                                                               | 30               |
| CO-4    | Develop GUI-based applications using JavaFX with proper event handling and layout management.                                                                                                                          | 10               |
| CO-5    | Utilize Java Collections and Generics to store, retrieve, and process data efficiently in Java programs.                                                                                                               | 10               |

## Teaching and Examination Scheme:

| Teaching - Learning scheme<br>(in Hours per Semester) |   |    |      |     | Total<br>Credits<br>=<br>TH/30 | Assessment Pattern and Marks |           |                      |            |         | Total<br>Marks |
|-------------------------------------------------------|---|----|------|-----|--------------------------------|------------------------------|-----------|----------------------|------------|---------|----------------|
| L                                                     | T | P  | PBL* | TH  |                                | Theory                       |           | Tutorial / Practical |            |         |                |
|                                                       |   |    |      |     |                                | ESE<br>(E)                   | PA<br>(M) | PA (I)               | PBL<br>(I) | ESE (V) |                |
| 45                                                    | 0 | 30 | 45   | 120 | 04                             | 70                           | 30        | 20                   | 30         | 50      | 200            |

\* Problem Based Learning (PBL) aims to accommodate learning beyond syllabus as per clause 9.4 of NBA manual.



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000231

Subject Name: Object Oriented Programming

## Course Content:

| Sr. No. | Content                                                                                                                                                                                                                                                                                                                                                                                                                                     | Total Hrs | % of Weightage |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|
| 1       | <b>Basic of Java:</b> Java Platform and Architecture (JVM, JDK, JRE), Java Program Structure, Compilation, and Execution Java program, Data types (byte, short, int, long, float, double, char, boolean, String, Arrays), variables, keywords, literal, operators, operator precedence and associativity, type conversion                                                                                                                   | 3         | 5              |
| 2       | <b>Conditional and looping statements:</b> simple <i>if</i> , <i>if-else</i> statement, <i>else-if</i> ladder, <i>switch-case</i> statement, <i>for</i> loop, <i>while</i> loop, <i>do-while</i> loop, Enhanced <i>for</i> Loop ( <i>for-each</i> loop), <i>break</i> , <i>continue</i> , labelled <i>break</i> , labelled <i>continue</i>                                                                                                  | 3         | 5              |
| 3       | <b>Basics of Object Oriented Programming:</b> Data Encapsulation by defining classes, data members, member functions, Data Abstraction, access specifiers, static data member and static member function, defining constructors, objects and references, <i>this</i> keyword, returning and passing objects, array of objects, static block, instance block, inner class                                                                    | 7         | 20             |
| 4       | <b>Inheritance, Polymorphism and Wrapper classes:</b> Inheritance, types of inheritance, <i>super</i> , <i>this</i> and <i>final</i> keywords, method overloading, method overriding, dynamic binding, casting objects, instanceof operator, protected data members, String, StringBuffer and StringBuilder class, Wrapper classes for Primitive types, autoboxing/unboxing, Collection framework, Java API Packages, User-defined packages | 8         | 20             |
| 5       | <b>Interface, Abstract class and Exception Handling:</b> Interface, Partial interface, Multiple inheritance using interface, Abstract class, Abstract methods, Handling Exception using <i>try</i> , <i>catch</i> and <i>finally</i> keywords, Throwing exception using <i>throw</i> , checked and unchecked exception, <i>throws</i> keyword, Creating user-defined exceptions                                                             | 7         | 20             |
| 6       | <b>Concurrency control:</b> Thread life cycle, Creating threads using <i>Thread</i> class and <i>Runnable</i> interface, Thread methods, Thread synchronization using <i>synchronized</i> keyword                                                                                                                                                                                                                                           | 5         | 10             |
| 7       | <b>I/O Management and Generics:</b> , <i>File</i> class, Writing, reading and Random Access files, Purpose of Generics, Defining Generic classes and methods, Generics with Collections like ArrayList, HashMap, etc                                                                                                                                                                                                                        | 6         | 10             |
| 8       | <b>Designing GUI Applications using JavaFx:</b> Basic structure of JavaFx application, Layout Panes, UI Components, <i>Color</i> and <i>Font</i> classes, Event Handling                                                                                                                                                                                                                                                                    | 6         | 10             |
|         |                                                                                                                                                                                                                                                                                                                                                                                                                                             | 45        | 100            |



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000231

Subject Name: Object Oriented Programming

## Suggested Specification table with Marks (Theory): (For B.E. only)

| Distribution of Theory Marks |         |         |         |         |         |
|------------------------------|---------|---------|---------|---------|---------|
| R Level                      | U Level | A Level | N Level | E Level | C Level |
| 10                           | 20      | 40      | 00      | 00      | 00      |

**R: Remembrance; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create and above Levels (Revised Bloom's Taxonomy)**

Note: This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary slightly from above table.

### Reference Books:

1. **Java: The Complete Reference** – Herbert Schildt, McGraw Hill Education
2. **Head First Java** – Kathy Sierra, Bert Bates, O'Reilly Media.
3. **Core Java: An Integrated Approach** – R. Nageswara Rao, Dreamtech Press
4. **Thinking in Java** – Bruce Eckel, Pearson Education
5. **Object-Oriented Programming with Java** – C. Thomas Wu, Tata McGraw Hill
6. **Programming with Java** – E. Balagurusamy, McGraw Hill Education

### Suggested MOOC Courses:

1. ([https://onlinecourses.nptel.ac.in/noc22\\_cs47/preview](https://onlinecourses.nptel.ac.in/noc22_cs47/preview))
2. Java by Prof. Kannan Moudgalya, Indian Institute of Technology Bombay  
([https://onlinecourses.swayam2.ac.in/aic20\\_sp13/preview](https://onlinecourses.swayam2.ac.in/aic20_sp13/preview))
3. Object oriented Programming in Java by Mrs. Priyanka Sharma, Manav Rachna International Institute of Research and Studies  
([https://online-degree.swayam2.ac.in/mri22\\_01\\_d03\\_s1\\_cc3/preview](https://online-degree.swayam2.ac.in/mri22_01_d03_s1_cc3/preview))

### List of Experiments:

| Sr. No                                  | Practical                                                                                                                                                                                                                                                                                                                | CO  |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Practical Set-1 (Basics of JAVA)</b> |                                                                                                                                                                                                                                                                                                                          | CO1 |
| 1.1                                     | Develop a Java program that prompts the user to enter a distance in meters. Your program should then convert this distance to feet (1 meter = 3.28084 feet) and display the result formatted to two decimal places.                                                                                                      |     |
| 1.2                                     | Write a Java program to solve a system of two linear equations with two variables (e.g., $ax + by = e$ and $cx + dy = f$ ). Prompt the user to enter the coefficients a, b, c, d, e, f. Calculate and display the values of x and y using Cramer's rule. Include error handling for cases where the denominator is zero. |     |



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000231

Subject Name: Object Oriented Programming

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|                                               | Cramer's rule : $D=ad-bc$ , $D_x=ed-bf$ , $D_y=af-ce$ , $x=D_x/D$ , $y=D_y/D$                                                                                                                                                                                                                                                                                                                                                                                                               |            |
| 1.3                                           | Write a Java program that prompts the user to enter a single letter (character). Determine whether the entered character is a vowel (a, e, i, o, u, case-insensitive) or a consonant, and display the result.                                                                                                                                                                                                                                                                               |            |
| 1.4                                           | Develop a Java application that calculates a person's Body Mass Index (BMI). The program should ask the user for their weight in pounds and height in inches. Convert these values to kilograms and meters respectively (1 pound = 0.45359237 kg, 1 inch = 0.0254 meters) and then calculate BMI (weight in kg / (height in meters) <sup>2</sup> ). Display the calculated BMI.                                                                                                             |            |
| 1.5                                           | Simulate a simple ATM or cashier. Given an integer amount to be dispensed (e.g., 787), calculate and display the minimum number of currency notes of denominations 100, 50, 10, 5, 2, and 1 that would be given to the user.                                                                                                                                                                                                                                                                |            |
| 1.6                                           | Write a Java program that accepts a five-digit integer from the keyboard. Your program should then create a new number by adding one to each digit of the input number. For example, if the input is 12391, the output should be 23402 (note: 9+1=10, so it becomes 0 with a carry)                                                                                                                                                                                                         |            |
| 1.7                                           | Write a program that takes the lengths of three sides of a triangle as input. Calculate and print the area of the triangle. Ensure that the program validates if the given side lengths can actually form a triangle (sum of any two sides must be greater than the third side) before calculating the area (use Heron's formula: $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a+b+c)/2$ ).                                                                                          |            |
| 1.8                                           | Write a Java program that accepts two numbers as command-line arguments. Convert these arguments to appropriate numeric types (e.g., int or double), perform a simple calculation (e.g., sum or product), and print the result to the console.                                                                                                                                                                                                                                              |            |
| <b>Practical Set-2 (Basic OOP concepts-1)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>CO2</b> |
| 2.1                                           | Define a Java class named <i>Rectangle</i> . It should have two double data fields: <i>width</i> and <i>height</i> , both with a default value of 1. Implement a no-argument constructor and a constructor that takes <i>width</i> and <i>height</i> as parameters. Include methods <i>getArea()</i> and <i>getPerimeter()</i> that return the calculated area and perimeter respectively.                                                                                                  |            |
| 2.2                                           | Create an <i>Employee</i> class with private instance variables for <i>employeeName</i> (String) and <i>employeeSalary</i> (double). Implement public methods <i>readEmployeeData()</i> (to take input from the user) and <i>displayEmployeeData()</i> (to print the employee's name and salary). Demonstrate object creation and method invocation in a <i>main</i> method.                                                                                                                |            |
| 2.3                                           | Design a class <i>Time</i> with <i>hours</i> (int) and <i>minutes</i> (int) as data members. Include method <i>setTime(int h, int m)</i> to initialize the time and <i>displayTime()</i> to display the time. Implement a method <i>addTime(Time t1, Time t2)</i> that takes two Time objects as arguments, adds their hours and minutes, and stores the result in the calling object. Do not use constructors for initialization in this specific practical.                               |            |
| 2.4                                           | Create a <i>Point</i> class representing a 2D point (x, y). Implement a default constructor that initializes both x and y to 5. Provide a parameterized constructor to initialize x and y with user-supplied values. Also, implement a copy constructor to create a new <i>Point</i> object as a copy of an existing Point object. Include a <i>display()</i> method to show the point's coordinates and write a <i>main</i> method to test all constructors and the display functionality. |            |
| 2.5                                           | Define a Java class named <i>Rectangle</i> . It should have two double data fields: <i>width</i> and                                                                                                                                                                                                                                                                                                                                                                                        |            |



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000231

Subject Name: Object Oriented Programming

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|                                               | <p>height. In your main method, create two Rectangle objects: one with <i>width</i> 4 and <i>height</i> 40, and another with <i>width</i> 3.5 and <i>height</i> 35.9. For each rectangle, display its <i>width</i>, <i>height</i>, calculated area, and perimeter. Then, compare the two rectangles based on their areas and print which one has a larger area.</p>                                                                                                                                                                                                                                                                                                                                                       |            |
| 2.6                                           | <p>Create a class <i>BankAccount</i> with <i>accountId</i> (String), <i>accountHolderName</i> (String), and <i>balance</i> (double) as instance variables. Include methods <i>assignValues()</i> (for initialization) and <i>displayValues()</i>. Implement a search function that takes an <i>accountId</i> as input and, if found within an array of <i>BankAccount</i> objects, displays the details of that specific account. In your main method, create an array of at least five <i>BankAccount</i> objects and demonstrate adding, displaying, and searching for accounts.</p>                                                                                                                                    |            |
| 2.7                                           | <p>Write a program for billing system for a shopping mall. Create a class <i>BillGenerator</i> that uses method overloading to generate bills based on customer type:<br/><i>generateBill(int itemTotal)</i>: For regular customers, apply no discount.<br/><i>generateBill(int itemTotal, int discount)</i>: For privileged customers, apply flat discount in rupees.<br/><i>generateBill(int itemTotal, double discountPercent)</i>: For festive offers, apply percentage discount.<br/>Write a program to display the final bill amount using appropriate overloaded method based on customer category.</p>                                                                                                            |            |
| 2.8                                           | <p>A bank wants to offer a facility to calculate EMI (Equated Monthly Installment) for different types of loans. Design a class <i>LoanCalculator</i> with the following overloaded methods:<br/><i>calculateEMI(int principal, int time, float rate)</i>: For home loans<br/><i>calculateEMI(double principal, int time, double rate)</i>: For vehicle loans<br/><i>calculateEMI(int principal, int time)</i>: For short-term personal loans with a fixed interest rate of 10%<br/>Demonstrate the use of all three methods in the main method by calculating EMIs for different loan types.</p>                                                                                                                         |            |
| <b>Practical Set-3 (Basic OOP concepts-1)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>CO2</b> |
| 3.1                                           | <p>Create a Java class named <i>University</i> with a static data member <i>totalStudents</i> to keep track of the number of student objects created. Implement a static method <i>getTotalStudents()</i>. Also, include a static block to initialize a static variable (e.g., <i>universityName</i>) and an instance block to print a message when an object is created. Demonstrate their execution order.</p>                                                                                                                                                                                                                                                                                                          |            |
| 3.2                                           | <p>Design a class <i>BankAccount</i> with <i>account_holder_name</i> and <i>balance</i>. Use a static variable <i>interest_rate</i> (same for all accounts). Include methods to calculate and display the interest earned. Update interest rate using a static method.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |
| 3.3                                           | <p>Write a Java program to model a college admission system using the concept of inner classes. Create an outer class named <i>College</i> that stores the <i>collegeName</i> as a data member and initializes it through a constructor. Within the <i>College</i> class, define a non-static inner class named <i>Admission</i> that contains student-specific details such as the <i>studentName</i> and the course they are enrolling in. The inner class should have methods to accept and display student information, and it should also be able to access and display the <i>collegeName</i> stored in the outer class. In the <i>main</i> method, create an object of the <i>College</i> class by passing the</p> |            |



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|                                                         | <i>collegeName</i> , and then use this object to create an instance of the inner <i>Admission</i> class. Invoke appropriate methods to input the student's name and course, and then display the complete admission details, including the college name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |
| 3.4                                                     | Create a Java method <i>isValidPassword</i> (String password) that checks if a given string is a valid password based on the following rules: <ul style="list-style-type: none"><li>• It must have at least eight characters.</li><li>• It must consist only of letters and digits.</li><li>• It must contain at least two digits.</li></ul> The program should prompt the user to enter a password and display "Valid Password" or "Invalid Password" accordingly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |
| 3.5                                                     | Write a Java program that demonstrates method overloading to calculate the volume of different 3D shapes. Implement overloaded methods named <i>calculateVolume()</i> for a Cube (takes one side length), a RectangularCube (takes length, width, height), and a Sphere (takes radius).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |
| <b>Practical Set-4 (Inheritance &amp; Polymorphism)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>CO2</b> |
| 4.1                                                     | Design a base class <i>Shape</i> with two double data members <i>d1</i> and <i>d2</i> to store dimensions. Include a method <i>getData</i> (double <i>d1</i> , double <i>d2</i> ) to initialize these dimensions. Create two derived classes, <i>Triangle</i> and <i>Rectangle</i> , which inherit from <i>Shape</i> . Each derived class should have its own method to calculate its specific area.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |
| 4.2                                                     | Define a base class <i>BankAccount</i> with common attributes like <i>accountNumber</i> , <i>accountHolderName</i> , and <i>balance</i> . Then, define two subclasses: <i>SavingAccount</i> and <i>FixedDepositAccount</i> , which inherit from <i>BankAccount</i> . Implement basic operations like <i>openAccount()</i> , <i>deposit()</i> , <i>checkBalance()</i> , and <i>withdraw()</i> in <i>BankAccount</i> . The <i>SavingAccount</i> should include a <i>calculateInterest()</i> method specific to savings accounts, and <i>FixedDepositAccount</i> should have a <i>maturityAmount()</i> method considering fixed deposit terms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |
| 4.3                                                     | Create a base class named <i>Employee</i> that contains a method <i>displayDetails()</i> which prints general employee details such as <i>name</i> and <i>department</i> . Now create a subclass <i>Manager</i> that inherits from <i>Employee</i> and overrides the <i>displayDetails()</i> method to include additional information such as the manager's team size or project name. In the <i>main</i> method, create objects of both <i>Employee</i> and <i>Manager</i> classes and call the <i>displayDetails()</i> method using each object to show how Java determines which version of the method to execute at runtime.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |
| 4.4                                                     | Create a base class named <i>Vehicle</i> that contains common attributes such as <i>vehicleNumber</i> , <i>brand</i> , and <i>fuelType</i> . Include a constructor to initialize these fields and a method <i>displayDetails()</i> to print them. Derive a subclass <i>Car</i> from <i>Vehicle</i> which adds attributes such as <i>numberOfSeats</i> and <i>ACavailable</i> (boolean). Override the <i>displayDetails()</i> method to include the car-specific details, and use the <i>super</i> keyword to invoke the parent class constructor and methods. Further, derive another subclass <i>ElectricCar</i> from <i>Car</i> that includes attributes such as <i>batteryCapacity</i> and <i>chargingTime</i> , and again override the <i>displayDetails()</i> method to include electric car-specific details. Demonstrate constructor chaining, method overriding, use of protected access specifier for inherited members, and <i>instanceof</i> operator to check object type at runtime. In the <i>main()</i> method, create objects of all three classes and display their details using overridden methods. Also, use upcasting (Vehicle v = new Car(...)) and downcasting with <i>instanceof</i> check to access subclass- |            |



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|                                                                | specific features.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |
| 4.5                                                            | Write a Java program to demonstrate method overriding in an online payment system. Create a superclass <i>Payment</i> with a method <i>processPayment(int amount)</i> . In the subclass <i>CreditCardPayment</i> , override the method to print "Payment of Rs.<amount> done using Credit Card". In the subclass <i>UPIPayment</i> , override the method to print "Payment of Rs.<amount> done using UPI". Accept user choice and amount, and display the payment result using method overriding.                                                                                                                                                                                                                                                                                      |            |
| <b>Practical Set-5 (Interface, Abstract Class and Package)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>CO2</b> |
| 5.1                                                            | Given an interface <i>Classify</i> with a method <i>String getDivision(double average)</i> . Implement this <i>getDivision</i> method in a class <i>Result</i> such that it returns "First Division" if the average is 60 or more.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |
| 5.2                                                            | Write the Java code for an interface named <i>Exam</i> which declares a single abstract method <i>boolean isPassed(int mark)</i> . This method should take an integer <i>mark</i> as an argument. Write the Java code for another interface named <i>Classify</i> which declares a single abstract method <i>String getDivision(double average)</i> . This method should take a double <i>average</i> as an argument. Create a class named <i>Result</i> that implements both the <i>Exam</i> and <i>Classify</i> interfaces. Provide concrete implementations for <i>isPassed()</i> and <i>getDivision()</i> methods. In your <i>main</i> method, create an instance of <i>Result</i> , set some <i>marks</i> and <i>average</i> , and demonstrate the use of both interface methods. |            |
| 5.3                                                            | Design a Java program for an Online Order Processing System using partial interface implementation. Create an interface <i>Order</i> with three methods:<br><i>placeOrder(String item, int qty)</i><br><i>cancelOrder(int orderId)</i><br><i>generateBill()</i><br>Create an abstract class <i>PartialOrder</i> that implements the <i>Order</i> interface but provides implementation only for <i>placeOrder()</i> (storing order details). Create a concrete class <i>FinalOrder</i> that extends <i>PartialOrder</i> and implements the remaining methods <i>cancelOrder()</i> and <i>generateBill()</i> . Accept user input for order details and allow user to either generate a bill or cancel the order.                                                                        |            |
| 5.4                                                            | Write a Java program to create an abstract class <i>Vehicle</i> with: <ul style="list-style-type: none"><li>• An abstract method <i>fuelType()</i> that returns the type of fuel used.</li><li>• An abstract method <i>noOfWheels()</i> that returns the number of wheels.</li></ul> Create two subclasses: <ul style="list-style-type: none"><li>• Car that uses Petrol/Diesel and has 4 wheels.</li><li>• Bike that uses Petrol and has 2 wheels.</li></ul> In the main method, create objects of Car and Bike, and display their fuel type and number of wheels.                                                                                                                                                                                                                    |            |
| 5.5                                                            | Write a Java program using packages to generate a mark sheet for students. Create a package <i>student</i> that contains a class <i>Student</i> with the following: <ul style="list-style-type: none"><li>• Data members: <i>rollNo</i>, <i>name</i>.</li><li>• A constructor to initialize student details.</li><li>• A method <i>displayStudent()</i> to display student information.</li></ul> Create another package <i>exam</i> that contains a class <i>Result</i> which: <ul style="list-style-type: none"><li>• Extends the <i>Student</i> class.</li></ul>                                                                                                                                                                                                                    |            |



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|                                             | <ul style="list-style-type: none"><li>Has data members: <i>marks1</i>, <i>marks2</i>, <i>marks3</i>.</li><li>A method <i>displayResult()</i> that prints the student's mark sheet including total and average marks.</li></ul> <p>In the main method (inside the exam package), create a student with marks and display the mark sheet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |
| 5.6                                         | <p>Write a Java program using four different packages to demonstrate the use of access specifiers.</p> <ul style="list-style-type: none"><li>Package <i>apack</i>:<ul style="list-style-type: none"><li>Define a class <i>A</i> with three variables:<ul style="list-style-type: none"><li><i>public int pubVar</i></li><li><i>protected int protVar</i></li><li><i>private int privVar</i></li></ul></li><li>Provide a constructor to initialize them.</li></ul></li><li>Package <i>bpack</i>:<ul style="list-style-type: none"><li>Define a class <i>B</i> that extends <i>A</i>.</li><li>Create a <i>display()</i> method that tries to access variables of <i>A</i> using inheritance.</li></ul></li><li>Package <i>cpack</i>:<ul style="list-style-type: none"><li>Define a class <i>C</i> with a <i>display()</i> method.</li><li>Inside <i>display()</i>, create an object of class <i>A</i> and try to access its variables.</li></ul></li><li>Package <i>dpack</i>:<ul style="list-style-type: none"><li>Define a class <i>ProtectedDemo</i> with <i>main()</i>.</li><li>Create objects of class <i>B</i> and class <i>C</i>.</li><li>Call their respective <i>display()</i> methods to show which variables are accessible and which are not.</li></ul></li></ul> |            |
| <b>Practical Set-6 (Exception Handling)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>CO3</b> |
| 6.1                                         | Take the value of denominator and numerator from user using command-line argument. Implement the concept of exception handling to manage all possible run-time error.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |
| 6.2                                         | <p>Write a Java program to create a class <i>VotingApp</i> where:</p> <ul style="list-style-type: none"><li>The method <i>checkEligibility(int age)</i> checks if a person is eligible to vote.</li><li>If <i>age &lt; 18</i>, explicitly throw the predefined exception <i>IllegalArgumentException</i> with the message "Age must be 18 or above to vote".</li></ul> <p>In the main method, test the method with different age inputs.</p> <ul style="list-style-type: none"><li>Use a try-catch-finally block to handle exceptions.</li><li>The finally block should always print "Validation process completed"</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |
| 6.3                                         | <p>Write a Java application that defines a method <i>average(String[] values)</i> which:</p> <ul style="list-style-type: none"><li>Accepts an array of strings as an argument.</li><li>Converts each string element into a double and computes the average.</li><li>If any array element is null, the method should throw a <i>NullPointerException</i>.</li><li>If any element is not a valid number (e.g., "abc"), it should throw a <i>NumberFormatException</i>.</li><li>Include the throws clause in the method declaration.</li><li>In the main method, demonstrate the working of this program with valid and invalid inputs using <i>try-catch-finally</i></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |



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| 6.4                                                        | <p>Write a Java program to create a Banking Application using classes and exception handling. Create a class <i>BankAccount</i> with:</p> <ul style="list-style-type: none"> <li>• A constructor to initialize the balance with 1000.00.</li> <li>• A method <i>deposit(double amount)</i> to add money to the account.</li> <li>• A method <i>withdraw(double amount)</i> that subtracts money from the account.</li> <li>• If withdrawal is more than the available balance, it should throw a custom exception called <i>NotSufficientFundException</i>.</li> </ul> <p>In the main method:</p> <ul style="list-style-type: none"> <li>• Deposit 1000.00.</li> <li>• Perform three withdrawals: 400.00, 300.00, and 500.00.</li> <li>• The last withdrawal should throw the exception with the message "Not Sufficient Fund".</li> </ul>                                         |            |
| 6.5                                                        | <p>Define a custom exception class <i>BookNotAvailableException</i> that extends <i>Exception</i>.</p> <ul style="list-style-type: none"> <li>• Create a class <i>Library</i> with:</li> <li>• An instance variable <i>availableBooks</i> (integer).</li> <li>• A method <i>issueBook(int count)</i> that: <ul style="list-style-type: none"> <li>○ If <i>count</i> <math>\leq</math> <i>availableBooks</i>, reduce the number of books and display "Book issued successfully".</li> <li>○ Otherwise, throw <i>BookNotAvailableException</i> with the message "Requested books not available".</li> </ul> </li> </ul> <p>In the <i>main()</i> method:</p> <ul style="list-style-type: none"> <li>• Initialize the library with 3 available books.</li> <li>• Try issuing 2 books (valid).</li> <li>• Then try issuing 2 more books (should throw the custom exception).</li> </ul> |            |
| <b>Practical Set – 7 (Concurrency Control – Threading)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>CO3</b> |
| 7.1                                                        | <p>Write a Java program that creates two threads:</p> <ul style="list-style-type: none"> <li>• First thread prints numbers from 1 to 10 at the interval of 1 second.</li> <li>• Second thread prints numbers from 11 to 20 at the interval of 500 ms.</li> </ul> <p>Run both threads and display the output.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |
| 7.2                                                        | <p>Write a Java program where Thread T1 prints 1 to 100, T2 prints 101 to 200 and T3 prints 201 to 300. Initiate execution of all the three threads but ensure that numbers should be printed sequentially.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |
| 7.3                                                        | <p>Write a Java program where two threads print multiplication tables (e.g., Table of 5 and Table of 7). Use a synchronized method so that table outputs do not mix and remain consistent.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |
| 7.4                                                        | <p>Write a Java program where:</p> <ul style="list-style-type: none"> <li>• <b>Thread 1</b> computes the sum of numbers from 1 to 1000.</li> <li>• <b>Thread 2</b> computes the sum of numbers from 1001 to 2000.</li> <li>• Both threads should run <b>in parallel</b>.</li> <li>• Each thread should <b>return its computed sum</b> to the main method (e.g., using a getter method or storing result in a shared variable).</li> <li>• The <b>main method</b> should wait for both threads to finish using <i>join()</i>, then combine the two partial sums and print the <b>final result</b>.</li> </ul>                                                                                                                                                                                                                                                                       |            |
| <b>Practical Set – 8 (File I/O)</b>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>CO3</b> |



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| 8.1                                                          | Write a program that will count the number of characters, words, and lines in a file. Words are separated by whitespace characters. The file name should be passed as a command-line argument.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |
| 8.2                                                          | Create a file named <i>students.txt</i> . Write at least three student records (roll number, name, marks) into the file. Read the file content and display all student records on the console. Handle exceptions like <i>IOException</i> properly using try-catch-finally.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |
| 8.3                                                          | Write a Java program that reads a text file named <i>data.txt</i> . The program should count and display: The total number of lines, The total number of words, The total number of characters (excluding spaces and newline characters), Use <i>FileReader</i> / <i>BufferedReader</i> for reading the file.<br>Handle exceptions like <i>FileNotFoundException</i> and <i>IOException</i> .                                                                                                                                                                                                                                                                                                                                                    |            |
| 8.4                                                          | Write a Java program that takes <b>file name(s) and commands from the command line arguments</b> and performs the following operations: <ul style="list-style-type: none"><li>• <b>Copy</b> a file from source to destination.</li><li>• <b>Delete</b> a given file.</li><li>• <b>Rename</b> a file.</li></ul> After performing the operation, the program should print the following <b>file properties</b> : <ul style="list-style-type: none"><li>• File name</li><li>• Absolute path</li><li>• File size (in bytes)</li><li>• Whether the file is readable/writable</li><li>• Last modified date</li></ul> Use <i>File</i> class methods for properties, and handle exceptions such as <i>IOException</i> and <i>FileNotFoundException</i> . |            |
| <b>Practical Set – 9 (Collection Framework and Generics)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>CO5</b> |
| 9.1                                                          | Write a Java program that uses an <i>ArrayList&lt;Integer&gt;</i> to store marks of students. Perform the following operations: <ul style="list-style-type: none"><li>• Add at least 5 marks.</li><li>• Display all marks.</li><li>• Find and display the highest and lowest marks using <i>Collections.max()</i> and <i>Collections.min()</i>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                        |            |
| 9.2                                                          | Write a program that accepts a sentence from the user and counts the frequency of each word using a <i>HashMap&lt;String, Integer&gt;</i> . Display the results in the format:<br><b>Input:</b> "Java is fun and Java is powerful"<br><b>Output:</b><br>Java -> 2<br>is -> 2<br>fun -> 1<br>and -> 1<br>powerful -> 1                                                                                                                                                                                                                                                                                                                                                                                                                            |            |
| 9.3                                                          | Write a Java program to simulate a Music Playlist using <i>LinkedList&lt;String&gt;</i> . Perform the following operations: <ol style="list-style-type: none"><li>1. Add songs to the playlist.</li><li>2. Display the full playlist.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |



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|                                     | 3. Play the first song (remove from front).<br>4. Skip the last song (remove from end).<br>5. Display the updated playlist after each operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |
| 9.4                                 | Create a generic class <i>Box&lt;T&gt;</i> with a method <i>addItem(T item)</i> that stores items in an <i>ArrayList&lt;T&gt;</i> . Demonstrate the class by: <ul style="list-style-type: none"> <li>• Creating a <i>Box&lt;String&gt;</i> for names.</li> <li>• Creating a <i>Box&lt;Integer&gt;</i> for roll numbers.</li> <li>• Display stored items for both.</li> </ul>                                                                                                                                                                                                                                                                                                                                            |            |
| 9.5                                 | Write a generic method <i>searchElement</i> that accepts a <i>LinkedList&lt;T&gt;</i> and an element T to search. Return true if the element exists, otherwise false. <ul style="list-style-type: none"> <li>• Test with <i>LinkedList&lt;Integer&gt;</i> for roll numbers.</li> <li>• Test with <i>LinkedList&lt;String&gt;</i> for names.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                  |            |
| 9.6                                 | Write a generic method <i>sortList(List&lt;T&gt; list)</i> that sorts elements of an <i>ArrayList&lt;T&gt;</i> , where T extends <i>Comparable&lt;T&gt;</i> . Demonstrate with: <ul style="list-style-type: none"> <li>• An <i>ArrayList&lt;Integer&gt;</i> of numbers.</li> <li>• An <i>ArrayList&lt;String&gt;</i> of names.</li> <li>• Display the list before and after sorting.</li> </ul>                                                                                                                                                                                                                                                                                                                         |            |
| <b>Practical Set – 10 (JAVA FX)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>CO4</b> |
| 10.1                                | Write a JavaFX program that displays five Text nodes vertically (stacked). For each Text: <ul style="list-style-type: none"> <li>• Use font Times New Roman, bold + italic, size 22 px.</li> <li>• Assign a random color and random opacity (between 0.3 and 1.0) to each text.</li> <li>• Center the texts horizontally in the window and add spacing between them.</li> </ul> UI / Classes to use: VBox, Text, Font, Color, Random.<br>On launch the window shows five vertically arranged lines (e.g., “Text 1”, ... “Text 5”) each with different color & opacity and same font style.                                                                                                                              |            |
| 10.2                                | Create a JavaFX app that shows a circle which continuously bounces (left↔right) inside the scene. Provide: <ul style="list-style-type: none"> <li>• A Start and Stop button.</li> <li>• A Slider to control the ball speed (min: slow, max: fast).</li> <li>• When window is resized, the ball should keep bouncing within new bounds.</li> </ul> UI / Classes to use: Pane or AnchorPane, Circle, TranslateTransition or AnimationTimer, Button, Slider.<br>Following behavior is expected: <ul style="list-style-type: none"> <li>• Press Start → ball moves horizontally and reverses on boundaries.</li> <li>• Move Slider while running → ball speed updates.</li> <li>• Press Stop → animation pauses.</li> </ul> |            |
| 10.3                                | Design a registration form UI with fields: <i>Roll_NO</i> (numeric), <i>Name</i> , <i>Age</i> (numeric), <i>Email</i> and a Submit button. Requirements: <ul style="list-style-type: none"> <li>• Validate inputs on submit:               <ul style="list-style-type: none"> <li>◦ <i>RollNo</i> and <i>Age</i> must be integers.</li> <li>◦ <i>Email</i> must contain @ and . (basic check).</li> </ul> </li> <li>• On success show a confirmation Alert with entered data.</li> <li>• On validation failure show an error alert describing the issue.</li> </ul> UI / Classes to use: GridPane, TextField, Button, Alert, FileChooser,                                                                               |            |



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000231

Subject Name: Object Oriented Programming

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|      | FileWriter/BufferedWriter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
| 10.4 | <p>Write a JavaFX program that displays a bar chart to represent the percentage distribution of overall grades using Rectangle shapes.</p> <ul style="list-style-type: none"><li>• Projects: 20%, displayed in Red</li><li>• Quizzes: 10%, displayed in Blue</li><li>• Midterm Exams: 30%, displayed in Green</li><li>• Final Exam: 40%, displayed in Orange</li></ul> <p>Requirements:</p> <ol style="list-style-type: none"><li>1. Each category should be displayed with a labeled bar.</li><li>2. Bars should be proportional in height to the percentage.</li><li>3. Use the Rectangle class to create the bars.</li><li>4. Display the labels (e.g., "Projects — 20%") under each bar.</li><li>5. Arrange the bars horizontally in the scene using an HBox or Pane.</li></ol> |  |

## Major Equipment:

## List of Open Source Software/learning website:

### IDEs for Core Java

1. Eclipse IDE – <https://eclipse.org/downloads/>
2. NetBeans IDE – <https://netbeans.apache.org/download/>
3. IntelliJ IDEA Community Edition – <https://www.jetbrains.com/idea/download/>
4. Visual Studio Code - <https://code.visualstudio.com/>

### Libraries & Tools (Core Java Utilities)

1. Apache Commons – <https://commons.apache.org/>
2. Gson (JSON by Google) – <https://github.com/google/gson>
3. JUnit (Testing) – <https://junit.org/junit5/>

### JavaFX & GUI

1. OpenJFX (JavaFX official) – <https://openjfx.io/>
2. Scene Builder (Gluon) – <https://gluonhq.com/products/scene-builder/>
3. ControlsFX – <https://github.com/controlsfx/controlsfx>

## • List of suggested activities for Problem Based Learning:

| Sl. No. | Name of the activity                                          | No. of hours                          | Evaluation Criteria                |
|---------|---------------------------------------------------------------|---------------------------------------|------------------------------------|
| 1       | Assignment writing. Numerical based assignment is preferable. | 5 assignments of 3h each. Total = 15h | Based on the assignment submitted. |



# GUJARAT TECHNOLOGICAL UNIVERSITY

**Program Name: Bachelor of Engineering**

**Level: UG**

**Subject Code: BE04000231**

**Subject Name: Object Oriented Programming**

|    |                                                                                                      |                                                                                            |                                                                                         |
|----|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 2  | Problem solving/Coding using C, C++, Python, SCILAB, MATLAB, MS-EXCEL or any other relevant software | 5 small coding-based problems of 3h each. Total = 15h                                      | Based on the coding solution submitted.                                                 |
| 3  | Technical Video based learning related to the subject                                                | Duration of video = 5h<br>Report preparation & Presentation = 10h<br>Total = 15h           | Report /presentation based on the video learning outcomes.                              |
| 4  | Discussion on research paper based on relevant subject                                               | 3 research paper = 15h                                                                     | Summarize research paper and evaluation critical parameters                             |
| 5  | Poster/chart/power point preparation on technical topics                                             | Duration = 10 h                                                                            | Based on poster/chart preparation and presentation skills                               |
| 6  | Application/Software development                                                                     | Duration = 15 h                                                                            | Depending on the complexity of the Application/Software                                 |
| 7  | Group Discussion on emerging/trending technical topics based on subject                              | Duration = 1 h each                                                                        | Based on performance in group discussion, technical depth, knowledge etc.               |
| 8  | Seminar / Presentation                                                                               | Duration for study and preparation=5h<br>Report writing=3h<br>Presentation=2h<br>Total=10h | Topic can be selected technical content beyond syllabus                                 |
| 9  | Real world case studies-based learning                                                               | Duration of data collection/study = 5h<br>Report preparation = 10h Total = 15h             | Based on in-depth study, technical depth, data collected, fact finding, etc.            |
| 10 | Working/non-working model on technical topics                                                        | Working = 12 h<br>Non- working = 8 h                                                       | Based on inter department/external evaluation                                           |
| 11 | Self-learning on-line course                                                                         | Minimum duration of the course should be 15h.                                              | Examination based assessment at the end of course. Based on the certificate produced.   |
| 12 | Complex problem solving                                                                              | Maximum 3 problem. Study of the problem and solution finding, Total = 15h                  | Based on the depth of the solution submitted.                                           |
| 13 | Industry/Research laboratory visit                                                                   | Visit = 5h, Report preparation = 5h Total = 10h                                            | Based on report submitted. Report should contain observations and calculations based on |



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000231

Subject Name: Object Oriented Programming

|    |                                                                                                                            |                                                                                                                     | industry/ lab data.                                    |
|----|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 14 | Videos on Industrial safety aspects based on subject                                                                       | Duration of video = 5h<br>Report preparation = 5h<br>Total = 10h                                                    | Based on quiz/report submitted                         |
| 15 | Industrial exposure for 2-3 days to observe and provide tentative solutions on society/environment /health/any other issue | Duration = 15 h for industrial exposure<br><br>Problem identification and tentative solution = 10 h<br>Total = 20 h | Based on evaluation of critical problems and solutions |

Note:

- All the suggested activity should be related to the subject.
- Min 3 activities must be carried out as per the availability of faculties and students.
- The number of hours is suggestive. Faculty can sub-divide the number of hours based on the activity. However, total number of hours is fixed.
- Rubrics for the evaluation can be prepared by the faculty.

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# GUJARAT TECHNOLOGICAL UNIVERSITY

**Program Name: Bachelor of Engineering**

**Level: UG**

**Subject Code: BE04000221**

**Subject Name: Operating Systems**

|                         |                          |
|-------------------------|--------------------------|
| w. e. f. Academic Year: | 2024-25                  |
| Semester:               | 4                        |
| Category of the Course: | Professional Core Course |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisite:</b> | Knowledge of C Programming , Data Structures and Computer Architecture with assembly programming                                                                                                                                                                                                                                                                                                      |
| <b>Rationale:</b>    | Operating System is a core subject of Computer Engineering/Information Technology. The study of Operating Systems provides in-depth understanding of Program Execution Environment and management of various types of hardware and software resources in Computer. It involves important aspects of understanding Security, Virtualization, Storage and Access of Data and Parallelism of Processing. |

## Course Outcomes:

| Sr. No. | CO statement                                                                                                              | Marks% weightage |
|---------|---------------------------------------------------------------------------------------------------------------------------|------------------|
| CO-1    | Students will be able to understand the underlying components and architecture of the OS.                                 | 25               |
| CO-2    | Students will be able to conceptualize the executing program – with Process & Thread management.                          | 20               |
| CO-3    | Students will be able to learn and simulate existing algorithms and mechanisms for Memory Management.                     | 20               |
| CO-4    | Students will be able to analyze and compare File management and I/O Management with different Operating Systems.         | 20               |
| CO-5    | Students will be able to understand Virtualization, Security, Parallelism available in various existing OS and future OS. | 15               |

## Teaching and Examination Scheme:

| Teaching - Learning Scheme (in Hours per Semester) |   |    |      |     | Total Credits = TH/30 | Assessment Pattern and Marks |        |                      |         |         | Total Marks |
|----------------------------------------------------|---|----|------|-----|-----------------------|------------------------------|--------|----------------------|---------|---------|-------------|
| L                                                  | T | P  | PBL* | TH  |                       | Theory                       |        | Tutorial / Practical |         |         |             |
|                                                    |   |    |      |     |                       | ESE (E)                      | PA (M) | PA (I)               | PBL (I) | ESE (V) |             |
| 45                                                 | 0 | 30 | 45   | 120 | 04                    | 70                           | 30     | 20                   | 30      | 50      | 200         |

- Problem Based Learning (PBL) aims to accommodate learning beyond syllabus as per clause 9.4 of NBA manual.



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000221

Subject Name: Operating Systems

## Content:

| Sr. No. | Content                                                                                                                                                                                                                                                                                                                                                                                               | Total Hrs |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1       | <b>Introduction:</b> Computer system overview, Architecture, Goals & Structures of O.S, Basic functions, Interaction of O.S. & hardware architecture, System calls, Batch, multiprogramming. Multitasking, time sharing, parallel, distributed & real-time O.S.                                                                                                                                       | 04        |
| 2       | <b>Process and Threads Management:</b> Process Concept, Process states, Process control, Threads, Uni-processor Scheduling: Types of scheduling: Preemptive, Non preemptive, Scheduling algorithms: FCFS, SJF, RR, Priority, Thread Scheduling, Real Time Scheduling. System calls like ps, fork, join, exec family, wait.                                                                            | 05        |
| 3       | <b>Concurrency &amp; Synchronization:</b> Principles of Concurrency, Mutual Exclusion: S/W approaches, H/W Support, Semaphores, Pipes, Message Passing, Signals, and Monitors.                                                                                                                                                                                                                        | 05        |
| 4       | <b>Inter Process Communication:</b><br>Race Conditions, Critical Section, Mutual Exclusion, Hardware Solution, Strict Alternation, Peterson's Solution, The Producer Consumer Problem, Semaphores, Event Counters, Monitors, Message Passing, Classical IPC Problems: Reader's & Writer Problem, Dining Philosopher Problem etc., Scheduling concept and Scheduling Algorithms.                       | 06        |
| 5       | <b>Deadlock:</b> Principles of Deadlock, Starvation, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, System calls.                                                                                                                                                                                                                                                                       | 03        |
| 6       | <b>Memory Management:</b><br>Memory Management requirements, Memory partitioning: Fixed and Variable Partitioning, Memory Allocation: Allocation Strategies (First Fit, Best Fit, and Worst Fit), Swapping, Paging and Fragmentation. Demand Paging, Security Issues.<br><b>Virtual Memory:</b> Concepts, VM management, Page Replacement Policies (FIFO, LRU, Optimal, Other Strategies), Thrashing. | 07        |
| 7       | <b>I/O Management &amp; Disk scheduling:</b><br>I/O Devices, Organization of I/O functions, Operating System Design issues, I/O Buffering, Disk Scheduling (FCFS, SCAN, C-SCAN, SSTF), RAID, Disk Cache.                                                                                                                                                                                              | 05        |
| 8       | <b>Unix/Linux Operating System:</b><br>Development Of Unix/Linux, Role & Function Of Kernel, System Calls, Elementary Linux command & Shell Programming, Directory Structure, System Administration                                                                                                                                                                                                   | 05        |



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000221

Subject Name: Operating Systems

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|              | Case study: Linux, Windows Operating System                                                                                                                                                                                                                                                                                                                                                                                                         |           |
| 9            | <b>Virtualization Concepts:</b><br>Virtual machines; supporting multiple operating systems simultaneously on a single hardware platform; running one operating system on top of another. True or pure virtualization.<br><b>Approaches to Virtualization:</b><br>Processor Issue, Memory Management, I/O Management, VMware ESXi, Microsoft Hyper-V and Xen Variants, Java VM, Linux VServer Virtual Machine Architecture, Android Virtual Machine. | 05        |
| <b>TOTAL</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>45</b> |

## Suggested Specification table with Marks (Theory): (For B.E. only)

| Distribution of Theory Marks |         |         |         |         |         |
|------------------------------|---------|---------|---------|---------|---------|
| R Level                      | U Level | A Level | N Level | E Level | C Level |
| 25                           | 25      | 20      | 10      | 10      | 10      |

**R: Remembrance; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create and above Levels (Revised Bloom's Taxonomy)**

Note: This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary slightly from above table.

### Reference Books:

1. Modern Operating Systems by Andrew S Tanenbaum, Prentice Hall India.
2. Operating System Concepts by Silberschatz, Peter B. Galvin and Greg Gagne, Wiley-Indian Edition.
3. Principles of Operating Systems by Naresh chauhan, Oxford Press.
4. Operating Systems by D.M. Dhamdhare, Tata McGraw Hill.
5. Operating Systems – Internals and Design Principles by William Stallings, Prentice Hall India.
6. UNIX Concepts and Applications – by Sumitabha Das, Tata McGraw Hill.
7. Unix Shell Programming – by Yashwant Kanetkar, BPB publications.

### • List of Experiments:

| Sr. No. | Title of Practical |
|---------|--------------------|
|         | Practical 1        |



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000221

Subject Name: Operating Systems

|     |                                                                                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1 | Introduction & Study of various features of O.S.                                                                                                                                        |
| 1.2 | Study of basic Linux commands.                                                                                                                                                          |
| 1.3 | Study of Advance commands and filters of Linux/UNIX.                                                                                                                                    |
|     | <b>Practical 2</b>                                                                                                                                                                      |
| 2.1 | Study of Shell with its environment, Scripting & Features.                                                                                                                              |
| 2.2 | Write a shell script to find the sum & factorial of a given number.                                                                                                                     |
| 2.3 | Write a shell script to find if a given number is prime or not?                                                                                                                         |
|     | <b>Practical 3</b>                                                                                                                                                                      |
| 3.1 | Write a shell script to find if the given string is Palindrom or not?                                                                                                                   |
| 3.2 | Write a shell script to make a sum of digits of a given number.                                                                                                                         |
| 3.3 | Write a shell script to read n numbers as command arguments and sort them in descending order.                                                                                          |
|     | <b>Practical 4</b>                                                                                                                                                                      |
| 4.1 | Write a C program to create exactly 'n' child processes with use of fork() system call in Linux.                                                                                        |
| 4.2 | Write a C program to distribute logic/work among 'n' child processes created by fork() system call in Linux.                                                                            |
| 4.3 | Write a C Program to create two child processes and run 'who' and 'ls' command in it with use of fork() and exec() system call.                                                         |
|     | <b>Practical 5</b>                                                                                                                                                                      |
| 5.1 | Write a C Program to create two threads for addition and subtraction of two large size array in parallel.                                                                               |
| 5.2 | Write a C Program to create child processes for addition and subtraction of two large size array in parallel. Compare the parallelism achieved by Thread model and child Process model. |
|     | <b>Practical 6</b>                                                                                                                                                                      |
| 6.1 | Write a C programs to demonstrate the use of shared memory variables.                                                                                                                   |
| 6.2 | Write a C program to demonstrate and use of semaphores in Linux.                                                                                                                        |
|     | <b>Practical 7</b>                                                                                                                                                                      |
| 7.1 | Write a shell script to find max, min , average of given set of numbers.                                                                                                                |
| 7.2 | Write a shell script to complete CD command which take one number as argument and moves that many levels up from the current directory.                                                 |
|     | <b>Practical 8</b>                                                                                                                                                                      |
| 8.1 | Write a shell script to mail a common message to all users who are currently logged in.                                                                                                 |



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Subject Code: BE04000221

Subject Name: Operating Systems

|      |                                                                                  |
|------|----------------------------------------------------------------------------------|
| 8.2  | Write a shell script to display 5 largest files in the current directory.        |
| 8.3  | Write a shell script to count how many files in a given directory.               |
|      | <b>Practical 9</b>                                                               |
| 9.1  | Write a C program to implement “ls” command in Linux.                            |
| 9.2  | Write a C program to implement “who” command in Linux.                           |
|      | <b>Practical 10</b>                                                              |
| 10.1 | Write a shell script to translate all characters in File to upper case.          |
| 10.2 | Write a shell script to change file extension (.txt) to (.doc) using for loop.   |
| 10.3 | Write a shell script to compare two files to check whether they are same or not. |

## Major Equipment:

### List of Open Source Software/learning website:

1. [www.nptel.ac.in](http://www.nptel.ac.in)

2. <https://www.tutorialspoint.com/>

3. <https://www.udemy.com/>

4. <https://www.coursera.org/>

5. [https://www.cse.iitb.ac.in/~mythili/teaching/cs347\\_autumn2016/index.html](https://www.cse.iitb.ac.in/~mythili/teaching/cs347_autumn2016/index.html)

### • List of suggested activities for Problem Based Learning:

| Sl. No. | Name of the activity                                                                                 | No. of hours                                                                     | Evaluation Criteria                                         |
|---------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1       | Assignment writing. Numerical based assignment is preferable.                                        | 5 assignments of 3h each. Total = 15h                                            | Based on the assignment submitted.                          |
| 2       | Problem solving/Coding using C, C++, Python, SCILAB, MATLAB, MS-EXCEL or any other relevant software | 5 small coding-based problems of 3h each. Total = 15h                            | Based on the coding solution submitted.                     |
| 3       | Technical Video based learning related to the subject                                                | Duration of video = 5h<br>Report preparation & Presentation = 10h<br>Total = 15h | Report /presentation based on the video learning outcomes.  |
| 4       | Discussion on research paper based on relevant subject                                               | 3 research paper = 15h                                                           | Summarize research paper and evaluation critical parameters |



# GUJARAT TECHNOLOGICAL UNIVERSITY

**Program Name: Bachelor of Engineering**

**Level: UG**

**Subject Code: BE04000221**

**Subject Name: Operating Systems**

|    |                                                                                                                            |                                                                                                              |                                                                                                             |
|----|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 5  | Poster/chart/power point preparation on technical topics                                                                   | Duration = 10 h                                                                                              | Based on poster/chart preparation and presentation skills                                                   |
| 6  | Application/Software development                                                                                           | Duration = 15 h                                                                                              | Depending on the complexity of the Application/Software                                                     |
| 7  | Group Discussion on emerging/trending technical topics based on subject                                                    | Duration = 1 h each                                                                                          | Based on performance in group discussion, technical depth, knowledge etc.                                   |
| 8  | Seminar / Presentation                                                                                                     | Duration for study and preparation=5h<br>Report writing=3h<br>Presentation=2h<br>Total=10h                   | Topic can be selected technical content beyond syllabus                                                     |
| 9  | Real world case studies-based learning                                                                                     | Duration of data collection/study = 5h<br>Report preparation = 10h<br>Total = 15h                            | Based on in-depth study, technical depth, data collected, fact finding, etc.                                |
| 10 | Working/non-working model on technical topics                                                                              | Working = 12 h<br>Non- working = 8 h                                                                         | Based on inter department/external evaluation                                                               |
| 11 | Self-learning on-line course                                                                                               | Minimum duration of the course should be 15h.                                                                | Examination based assessment at the end of course. Based on the certificate produced.                       |
| 12 | Complex problem solving                                                                                                    | Maximum 3 problem. Study of the problem and solution finding, Total = 15h                                    | Based on the depth of the solution submitted.                                                               |
| 13 | Industry/Research laboratory visit                                                                                         | Visit = 5h, Report preparation = 5h<br>Total = 10h                                                           | Based on report submitted. Report should contain observations and calculations based on industry/ lab data. |
| 14 | Videos on Industrial safety aspects based on subject                                                                       | Duration of video = 5h<br>Report preparation = 5h<br>Total = 10h                                             | Based on quiz/report submitted                                                                              |
| 15 | Industrial exposure for 2-3 days to observe and provide tentative solutions on society/environment /health/any other issue | Duration = 15 h for industrial exposure Problem identification and tentative solution = 10 h<br>Total = 20 h | Based on evaluation of critical problems and solutions                                                      |



# **GUJARAT TECHNOLOGICAL UNIVERSITY**

**Program Name: Bachelor of Engineering**

**Level: UG**

**Subject Code: BE04000221**

**Subject Name: Operating Systems**

Note:

- All the suggested activity should be related to the subject.
- Min 3 activities must be carried out as per the availability of faculties and students.
- The number of hours is suggestive. Faculty can sub-divide the number of hours based on the activity. However, total number of hours is fixed.
- Rubrics for the evaluation can be prepared by the faculty.

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# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Branch: ALL Branches

Subject Code: BE04000101

Subject Name: Environmental Science, Sustainability and Renewable Energy

|                         |             |
|-------------------------|-------------|
| w. e. f. Academic Year: | 2024-25     |
| Semester:               | 4           |
| Category of the Course: | Core Course |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisite:</b> | Interest in natural systems sustaining life on the earth.                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Rationale:</b>    | This course aims to build environmental awareness and promote sustainable thinking among engineering students. It covers core environmental issues such as pollution, waste management, and climate change, while linking them to global challenges, need of renewable energy and sustainable development goals (SDGs). Students learn the scientific basis of environmental degradation and the role of engineering solutions in addressing them. |

## Course Outcomes:

| Sr. No. | CO statement                                                                                                                                           | Marks% weightage |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| CO-1    | Highlight the importance of environmental sciences.                                                                                                    | 5%               |
| CO-2    | Identify the types of pollution in society along with their sources, causes, effects and mitigation                                                    | 20%              |
| CO-3    | Explain the generation, impacts, and management of various types of wastes and describe the causes and effects of acid rain and ozone layer depletion. | 15%              |
| CO-4    | Describe the concepts of sustainability, climate change phenomena and green building principles.                                                       | 30%              |
| CO-5    | Recognize the role of Renewable Energy in sustainable development.                                                                                     | 30%              |

## Teaching and Examination Scheme:

| Teaching / Learning Scheme<br>(in Hours per semester) |    |   |      |                                      | Total<br>Credits | Assessment Pattern and Marks |                   |                      |            |            | Total<br>Marks |
|-------------------------------------------------------|----|---|------|--------------------------------------|------------------|------------------------------|-------------------|----------------------|------------|------------|----------------|
| L                                                     | T  | P | PBL* | Total no of<br>hours per<br>semester |                  | Theory                       |                   | Tutorial / Practical |            |            |                |
|                                                       |    |   |      |                                      |                  | ESE<br>(E)                   | PA /<br>CA<br>(M) | PA/<br>CA<br>(I)     | PBL<br>(I) | ESE<br>(V) |                |
| 30                                                    | 30 | 0 | 0    | 60                                   | 2                | 70                           | 30                | 20                   | 0          | 0          | 120            |

## Content:

| Sr. No. | Content                                                                                                                                                                                                                                                                                                             | Total Hrs |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1       | <b>INTRODUCTION TO ENVIRONMENT</b> <ul style="list-style-type: none"><li>Definition, principles and scope of Environmental Science. Impacts of technology on Environment, Environmental Degradation, Importance of environmental awareness and its conservation across different engineering disciplines.</li></ul> | 02        |



# GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Bachelor of Engineering

Level: UG

Branch: ALL Branches

Subject Code: BE04000101

Subject Name: Environmental Science, Sustainability and Renewable Energy

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |
| 2            | <b>ENVIRONMENTAL POLLUTION</b> <ul style="list-style-type: none"><li>• Water Pollution: Introduction – Water Quality Standards, Sources of Water Pollution, Classification of water pollutants, Effects of water pollutants</li><li>• Air Pollution: Composition of air, Structure of atmosphere, Ambient Air Quality Standards, Classification of air pollutants, Sources of common air pollutants like PM, SO<sub>2</sub>, NO<sub>x</sub>, Auto exhaust, Effects of common air pollutants</li><li>• Noise Pollution: Introduction, Sound and Noise, Noise measurements, Causes and Effects</li><li>• Land Pollution: Sources, causes, effects, control &amp; Prevention</li><li>• Solid Waste: Generation and management</li><li>• Bio-medical Waste: Generation and management</li><li>• E-waste: Generation and management</li><li>• Acid Rain, Depletion of Ozone layer and its mitigation.</li></ul> | 10          |
| 3            | <b>SUSTAINABILITY:</b> <ul style="list-style-type: none"><li>• Definition, scope of sustainability,</li><li>• Sustainable development &amp; Circular economy, Sustainable Development goals,</li><li>• Climate change: Global Warming and Green House Effect: concept and mechanism</li><li>• Climate Change Effects and Mitigation Strategies</li><li>• Green Building: Concept, Objectives, Core Principles &amp; Benefits.</li><li>• Concept of 4R's: Principles, Practical Application of 4R's.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                              | 09          |
| 4            | <b>RENEWABLE ENERGY:</b> <ul style="list-style-type: none"><li>• Conventional Vs Renewable Energy, Need of renewable energy</li><li>• Advantages, limitations and principles of generation of solar energy, wind energy, hydropower, biomass energy, geothermal energy, tidal energy</li><li>• Green Hydrogen: Concept, production, storage, transportation and application</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 09          |
| <b>TOTAL</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>100%</b> |

**Suggested Specification table with Marks (Theory): (For B.E. only)**

| Distribution of Theory Marks |         |         |         |         |         |
|------------------------------|---------|---------|---------|---------|---------|
| R Level                      | U Level | A Level | N Level | E Level | C Level |
| 40                           | 40      | 20      | 00      | 00      | 00      |



# GUJARAT TECHNOLOGICAL UNIVERSITY

**Program Name: Bachelor of Engineering**

**Level: UG**

**Branch: ALL Branches**

**Subject Code: BE04000101**

**Subject Name: Environmental Science, Sustainability and Renewable Energy**

**R: Remembrance; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create and above Levels (Revised Bloom's Taxonomy)**

Note: This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary slightly from above table.

## Reference Books:

1. Basics of Environmental Studies by Prof Dr N S Varandani ,2013 Publisher: LAP - LambertAcademic Publishing , Germany
2. Environmental Studies by Anindita Basak ,2009 Publisher: Drling Kindersley(India)Pvt. LtdPearson
3. Textbook of Environmental Studies by Deeksha Dave & S S Kateva , Cengage Publishers.
4. Environmental Sciences by Daniel B Botkin & Edward A Keller Publisher: John Wiley & Sons.
5. Environmental Studies by R. Rajagopalan, Oxford University Press
6. Environmental Studies by Benny Joseph, TMH publishers
7. Environmental Studies by Dr. Suresh K Dhameja, 2007 Published by : S K Kataria & SonsNew Delhi
8. Basics of Environmental Studies by U K Khare, 2011 Published by Tata McGraw Hill
9. Marco Alvera : The Hydrogen Revolution: a blueprint for the future of clean energy
10. The Energy and Resources Institute (TERI). (2022). *Hydrogen: India's Green Fuel of the Future*. New Delhi: TERI.

## List of Tutorials : Based on

1. Introduction to Environment
2. Water Pollution
3. Air Pollution
4. Noise Pollution
5. Solid Waste
6. Bio-medical Waste
7. E-waste
8. Global Environmental Issue
9. Concept of Sustainability
10. Renewable Energy

**List of Open Source Software/learning website : MOEF, NPTEL**

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